CIS 6600: Advanced Topics in Computer Graphics and Animation

Homework Assignment 4 (Spring 2026)

L-System Plug-in with Houdini Development Kit

Due: Monday, Feb. 23, 2026

The goal of this assignment is to learn how to use the Houdini Development Kit (HDK) to create C++ plugins for Houdini. In this assignment, the same LSystem.cpp file you used in the development of the Maya Plugin will be used in the creation of the Houdini plugin.

1. Setting up Visual Studio for Houdini (20 points)

If you haven't done so already, download the Houdini Apprentice Installer from https://www.sidefx.com/products/houdini-apprentice/ and install Houdini on your machine. The HDK is included with the Houdini install and is usually configured for the version of Visual Studio that was used to build Houdini. For example, for Houdini 21.0 the last 5 characters of the installer ("vc143") means it was built using Visual Studio 2022. As a result, you are recommended to use Visual Studio 2022 and Houdini 21.0 when developing your Houdini plugin for this assignment.

Note: Most of the Visual Studio settings are automatically setup in the project files provided with this assignment. Before starting, make sure that you set the following Windows environment variables: H21_PATH, H21_VERSION, CUSTOM_DSO_PATH, HOUDINI_DSO_PATH. Follow the instructions in **Appendix A** to set up your configurations.

For more details on the Houdini HDK and creating custom plugins see the Getting Started documentation at:
https://www.sidefx.com/docs/hdk/_h_d_k__intro__getting_started.html

2. Creating the Houdini L-System Node (80 points)

Once you have your development environment setup and working properly, the next step is to create a custom node using the Houdini Development Kit. The functionality of your Houdini node should be similar to the one that was created for the Maya plugin. This can be accomplished by following the instructions contained in the LsystemPlugin.c file of your Houdini Visual Studio project and completing all sections of the code where it says:

"PUT YOUR CODE HERE".

Note: The Houdini Development Kit (HDK) comes with a lot of examples located in the folder: C:\Program Files\Side Effects Software\Houdini 21.0. 596 \toolkit\samples

For this assignment, all the examples in the SOP folder, especially SOP_star, and SOP_PointWave are very helpful.

3. Extra credit (Maximum 50 points)

By default Houdini comes with an L-System node. Play around with the features of the default Houdini L-System node and compare it with the custom node you created. You will see a number of helpful features and options implemented inside the default L-System node. Try to recreate those features and options and incorporate them into your custom L-System node. You will receive 10 points for each new option and/or feature created, up to a maximum of 50 additional points.

Appendix A:

Setting up the HDK for Houdini 21.0 with Visual Studio 2022

1. Download and install Houdini 21.0 and Visual Studio 2022

2. Setup the following Windows Environment Variables

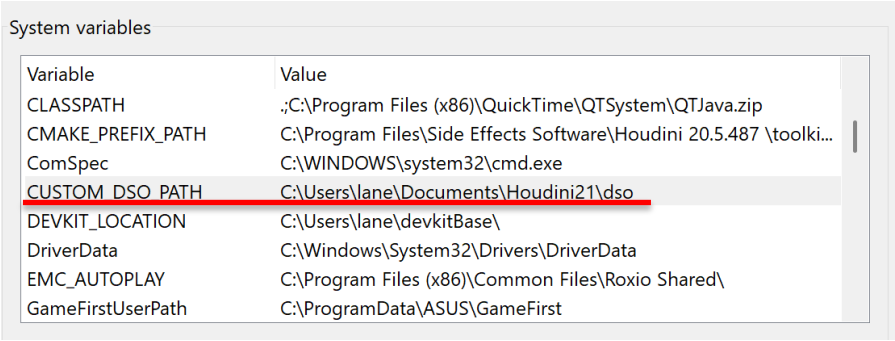
In Windows 11 you can modify Environment Variables by opening

Settings->System->About->Advanced System Settings->Environment Variables

The environment variables you need to set up are as follows:

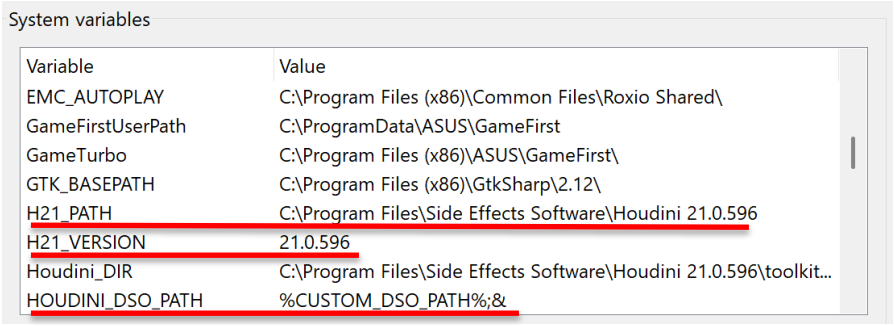
- H21_PATH: C:\Program Files\Side Effects Software\Houdini %H21_VERSION% (i.e. the path to where Houdini is installed)
- H21_VERSION: 21.0.596 (i.e. this should correspond to the version of Houdini you are actually using)
- CUSTOM_DSO_PATH: C:\Users**YOUR_USERNAME**\Documents\Houdini21\dso (i.e. this should represent the path to where your Houdini plugin dsos will be stored)
- HOUDINI_DSO_PATH: %CUSTOM_DSO_PATH%;& (extends the default dso path)

After this step, you should have following four variables:



System variables

Variable	Value
CLASSPATH	.;C:\Program Files (x86)\QuickTime\QTSystem\QTJava.zip
CMAKE_PREFIX_PATH	C:\Program Files\Side Effects Software\Houdini 20.5.487 \toolki...
ComSpec	C:\WINDOWS\system32\cmd.exe
CUSTOM_DSO_PATH	C:\Users\lane\Documents\Houdini21\dso
DEVKIT_LOCATION	C:\Users\lane\devkitBase\
DriverData	C:\Windows\System32\Drivers\DriverData
EMC_AUTOPLAY	C:\Program Files (x86)\Common Files\Roxio Shared\
GameFirstUserPath	C:\ProgramData\ASUS\GameFirst



System variables

Variable	Value
EMC_AUTOPLAY	C:\Program Files (x86)\Common Files\Roxio Shared\
GameFirstUserPath	C:\ProgramData\ASUS\GameFirst
GameTurbo	C:\Program Files (x86)\ASUS\GameFirst\
GTK_BASEPATH	C:\Program Files (x86)\GtkSharp\2.12\
H21_PATH	C:\Program Files\Side Effects Software\Houdini 21.0.596
H21_VERSION	21.0.596
Houdini_DIR	C:\Program Files\Side Effects Software\Houdini 21.0.596\toolkit...
HOUDINI_DSO_PATH	%CUSTOM_DSO_PATH%;&

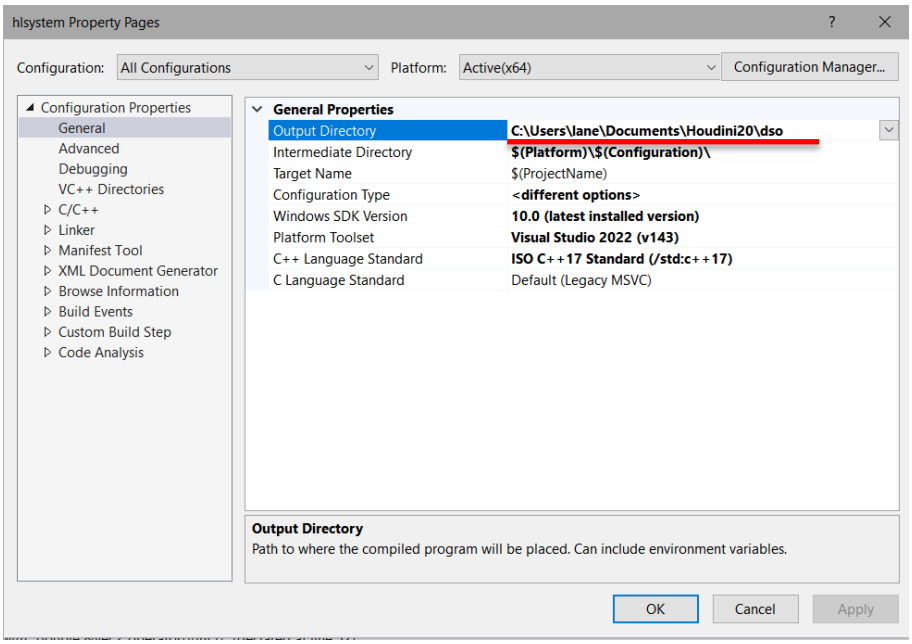
3. Download the file "Houdini Plugin Framework.zip file" from the CIS6600 Canvas site

- Unzip the file to your project folder
- Use CMake 3.24 or above to configure and generate the Visual Studio solution as follows:
    1. Open the Windows command prompt window
    2. In the command prompt window change to the directory where you have downloaded the Houdini Plugin code framework basecode (i.e. the "basecode" folder where you have the CMakeLists.txt file)
    3. At the Windows command prompt run the command

        cmake -G "Visual Studio 17 2022" -A x64 -S .\ -B .\Build

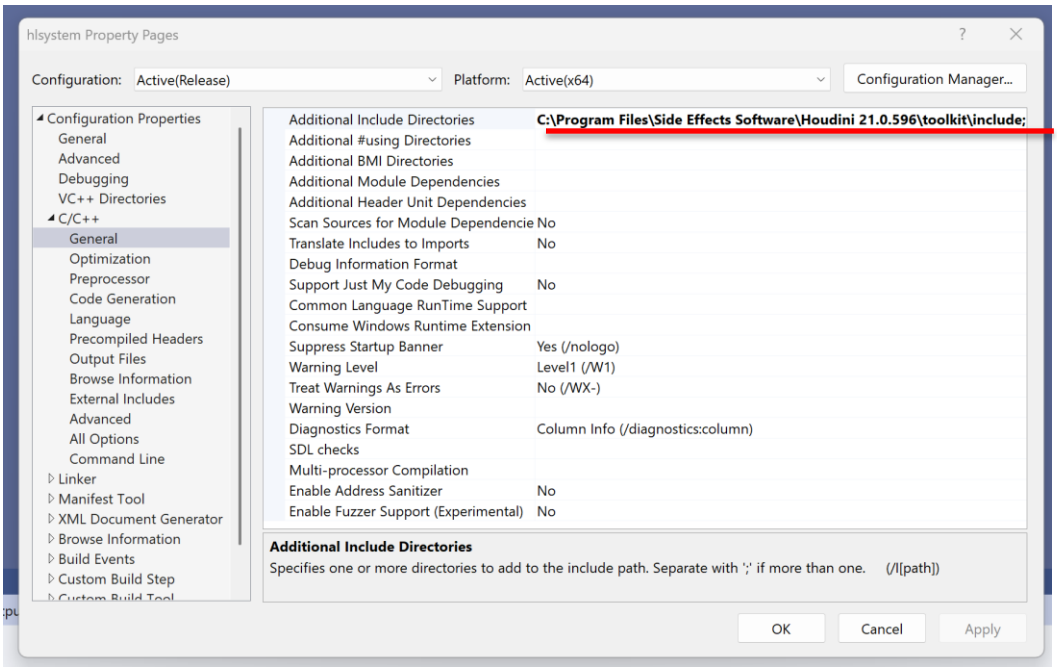
4. Using Windows FileExplorer go to the Houdini Plugin framework "Build" folder, double click on the VS solution file "hlsystem.sln"

- AfterbVisual Studio loads, right click on the hlsystem project and select "Properties"**.**
- Change the path in the Output Directory (for example, C:\Users\lane\Documents\Houdini21\dso) to the path where your Houdini plugin dsos will be stored



- Make sure the build configuration is set to **Release64** and **x64**.

In C++->General check the Additional Include Directory path properties to make sure it corresponds to where the Houdini toolkit include directory is located on your computer:

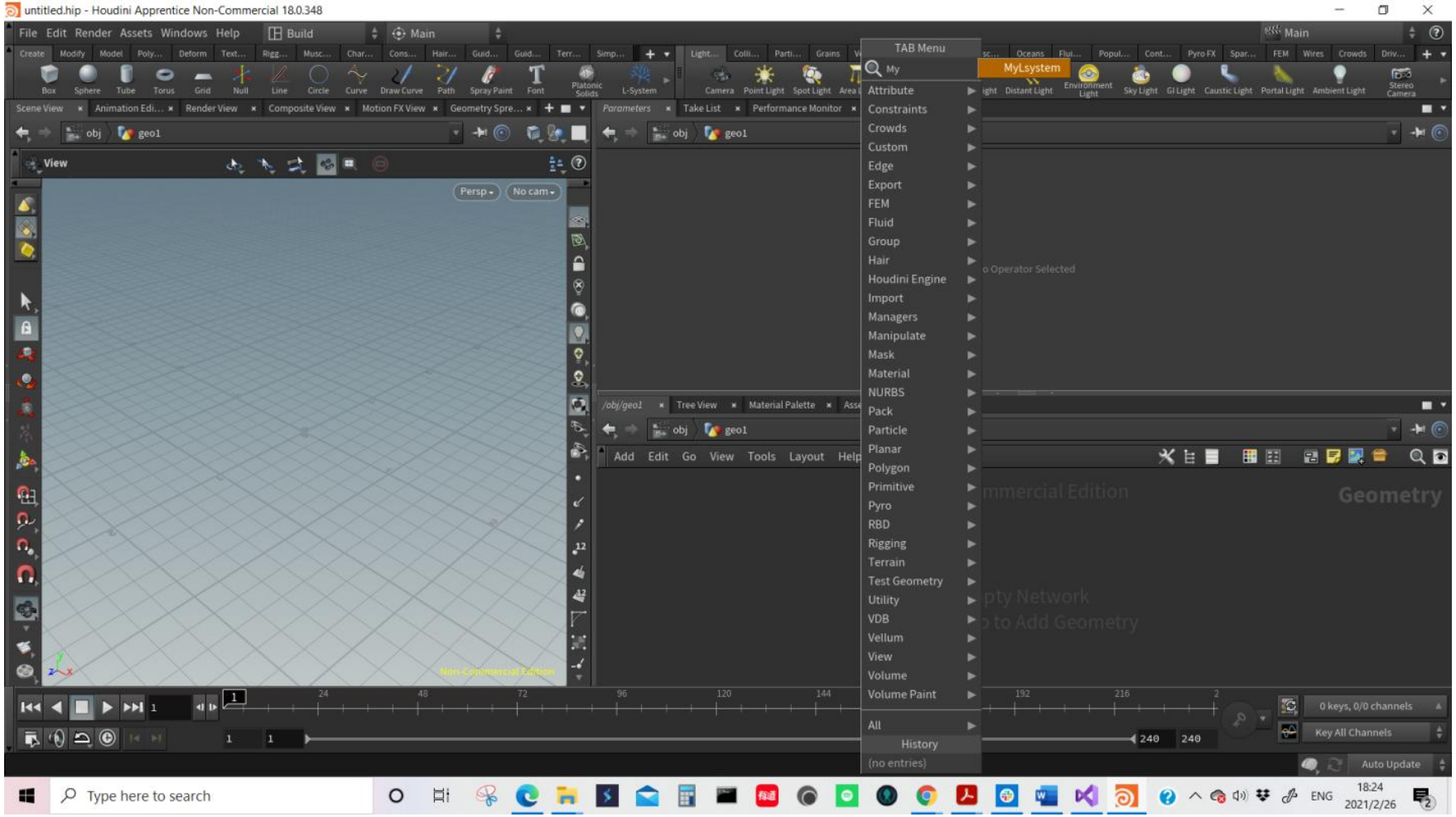


Loading your Houdini Plugin

1. **How to load your custom Houdini plugin?**
   - Move the cursor to the lower right panel in Houdini (Network Editor) and press the Tab key
   - Type "Geometry" then choose the geometry object.
   - Click to place in the Network Editor window.
   - Double-click the new geometry node to go into it.
   - Press Tab again in the Network Editor window and type "MyLsystem" (your plugin node).
   - Click to place. You should see inputs listed in the property window (upper-right panel). Changing these inputs should dynamically update your Lsystem.

2. **Reloading your plugin**
   - After recompiling your project you need to to close Houdini and reopen it again before in order to reload the plugin. Otherwise, you will see a compilation error.

3. **Extra Notes**
   - You need to use VS2022 to create plugins for Houdini 21.0. If you are using another version of Visual Studio (or Houdini) it will not work.
   - Houdini loads plugins in its default search path. For more details, please check this: https://www.sidefx.com/docs/hdk/_h_d_k__intro__creating_plugins.html
   - The basic plugin project should give you a SOP node called MyLsystem that can be found in the picture below.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Pyro
RBD
Rigging
Terrain
Test Geometry
Utility
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Vellum
View
Volume
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History
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Scene View
Animation Edi...
Render View
Composite View
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